

2026 ENVR Poster Sessions

Session 1

Isaac Amouzou - *Texas A&M*

Monte Carlo Dropout Hyperparameter Subgrids for Uncertainty Quantification in Spatial Deep Learning

Souvik Bag - *University of Missouri*

SURE Constrained Bayesian Models in the M-Open Setting

Souvick Bera - *Colorado School of Mines*

Distributionally Constrained Frequency Domain Empirical Likelihood

Srijato Bhattacharyya - *Texas A&M University*

Consistent Spatial Clustering of Complex Objects via Bregman Geometry

Susmit Bhattacharyya - *COLORADO SCHOOL OF MINES*

Marginal Distribution of the LatticeKrig coefficients under Spatial Autoregressive model with Spatial Extremes

Bianca Brusco - *University of California Irvine*

On Preferential Sampling for Spatial Extremes

Alejandro Calle-Saldaña - *UW-Madison*

From Multi-Scale Bayesian Transport Maps to Geometry-Aware Sparse Transformers for Spatial Fields

Linh Cao - *University of Houston*

Covariate-Assisted Latent Space Model

RJ Cass - *Brigham Young University*

A Simulation Based Approach for Exploring the Change of Support Problem and Related Questions

Christopher Chukwuemeka - *Tufts University*

Multivariate Spatio-Temporal Neural Hawkes Processes

Kevin Collins - *North Carolina State University*

Hierarchical Bayesian modeling for degraded point processes

Debangana Dey - *Texas A&M University*

Bigraphical Matern-Whittle (BMW) processes for fast inference of big multivariate spatial data on general domains

Thomas Hsiao - *Emory University*

Causal Sufficient Dimension Reduction for Multiple Continuous Exposures with an Application to Environmental Mixtures

Owen Jacobs - *North Carolina State University*

Investigating the Limits of Changepoint Detection for Spatio-Temporal Point Processes

Yeseul Jeon - *The University of North Carolina at Charlotte*

Deep Probabilistic Spatial Modeling for Multivariate Mixed-Type Responses

Jacob Johnson - *University of Wisconsin - Madison*

A Variational Ensemble Transport Filter for Nonlinear Data Assimilation

Niladri Kal - *Texas A&M University*

Uncertainty Quantification for Diffusion Surrogates in Spatial Prediction

Junhyeon Kwon - *University of North Texas*

Low-Rank Spatio-Temporal Hawkes Processes with Active-Set Selection

Isa Marques - *The Ohio State University*

Bayesian spatial+: a joint model perspective

Connor McNeill - *North Carolina State University*

Mitigating Spatial Confounding: An Assumption-Learn Approach

Ritwick Mondal - *Texas A&M University, College Station*

Bayesian Variable Selection Using Data Sketching for Spatially Varying Coefficient Models

Veronica Patterson - *Rice University*

Bayesian Network-on-Network (NoN) Regression for Edge-Level Prediction

Mary Salvana - *University of Connecticut*

Adaptive Block Mixed-Precision Cholesky Factorization for Large-Scale Space-Time Gaussian Processes

Di Shi - *University of Houston*

Spatial Inference from Aggregated Counts with Applications to Geolocation Error in Political Events

Muyang Shi - *University of California Santa Cruz*

Log-Laplace Nuggets for Fully Bayesian Fitting of Spatial Extremes Models to Threshold Exceedances

Landon Summers - *University of Arkansas*

HMMVI: an R library for modeling spatiotemporal precipitation data

Renhao Sun - *University of Texas at San Antonio*

Scalable Default Bayesian Analysis of a Multivariate Random Field

Paul Wiemann - *Ohio State University*

Scalable Variational Inference for Non-Gaussian Latent Fields via Sparse Autoregressive Flows

Troy Wixson - *The University of Massachusetts Amherst*

Tail Pairwise Dependence in R: from PCA to Time Series

Drew Yarger - *Purdue University*

Connecting reflective asymmetries in multivariate spatial and spatio-temporal covariances

Cong Ye - *Iowa State University*

A Bayesian Vecchia Gaussian Process Tree Model for High-Dimensional Data

Qishi Zhan - *Marquette University*

Bayesian Global-Deviation Spike-and-Slab Group Lasso for Spatially Varying Coefficient Models

Sherry Zhang - *University of Texas at Austin*

Creating glyph-maps for spatio-temporal data

Session 2

Megan Ayers - *Reed College*

Challenges and approaches for power analysis under interference: applications to PES program evaluation

Anik Burman - *Johns Hopkins University*

One Source, Many Monitors: Causal Machine Learning for Shared Environmental Exposures

Tony Cao - *Texas A&M University*

Characterizing Home Range and Spatial Use of White-tailed Hawks Along the Texas Gulf Coast (tentative)

Cesar Cardenas - *Rice University*

Mapping the Triple-demic: Clustering Texas Wastewater Treatment Plants via SARS-CoV-2, Influenza A, and RSV Dynamics

Adira Cohen - *North Carolina State University*

Unlocking large-scale abundance modeling through the fusion of human and AI annotations

Saleh Dadjouy - *University of Northern Colorado*

Weighting Geometry in Aggregate-Instrument Causal Analysis of Renewable Energy Policy and Emissions

James Doss-Gollin - *Rice University*

STAMPEDE: A Duration-Dependent Hierarchical Bayesian Framework for Nonstationary IDF Curves with Improved Out-of-Sample Sub-Hourly Estimates

Rajitha Thushan Edirisingha Edirisingha Mudiyansele - *Texas Tech University*

Bayesian Spatial Causal Modeling of Asthma Prevalence in the United States

Wenyu Gao - *UNC Charlotte*

Functional Clustering of Bird Migration Patterns Using Timing and Shape Information from the Motus Network

Wenlong Gong - *University of Houston - Downtown*

A nonstationary spatial model of PM2.5 with localized transfer learning from numerical model output

Madison Griffin - *William & Mary's Virginia Institute of Marine Science*

Losing Their Muscle?: A Bayesian Approach to Diagnosing the Effect of Acidic Events on Length-Weight Relationships for Atlantic Sea Scallops (*Placopecten magellanicus*)

Steven Hammond - <i>Montana State University</i> Reconstructing Environmental Systems from Irregular Observations with CRU-SHRED
Md Mahedi Hasan - <i>University of Wisconsin-Madison</i> Latent Representations for Dynamic Contact Networks in Wildlife Health Modeling and Uncertainty Quantification
Dharma Hoy - <i>Colorado State University</i> An Analysis of Heavy Rainfall with Extremal Imputation
Whitney Huang - <i>Clemson University</i> The History of ENVR: Celebrating the Past • Connecting the Present • Shaping the Future
Bora Jin - <i>Colorado School of Mines</i> Geometric source apportionment for Greenland Summit aerosol
Nicholas John - <i>University of Colorado Boulder</i> Correcting Bias in Machine Learning Weather Models
Katherine Kreuser - <i>Los Alamos National Laboratory</i> Physical-Statistical Modeling for Estimating Storm Surge r-Year Return Levels
Ryan Krumpelman - <i>Brigham Young University</i> Bayesian modeling of paleoclimate using proxy data sources
Konstantin Larin - <i>Rice University</i> Efficient Bayesian inference for two-stage models in environmental epidemiology
Nian Liu - <i>University of Evansville</i> A Neural-Mean Vecchia Gaussian Process for Unified Argo Modeling
Benjamin Lombardi - <i>CU Boulder</i> Spatial Clustering of Mesoscale Ocean Eddy Characteristics
Isaac MacArthur - <i>University of Massachusetts Amherst</i> Hierarchical Multinomial Logistic Regression (HMLR) for Forecasting SARS-CoV-2 Variants
Reetam Majumder - <i>University of Arkansas</i> Spatial Extremes at Scale: A Case Study of Surface Skin Temperature and Heat Risk in the United States
Camila Nunez Polanco - <i>University of Massachusetts, Amherst, NASA JPL</i> Honey, I Shrunk the Domain: Local Transport Maps for Spatial Aerosols
Anis Pakrashi - <i>The Pennsylvania State University</i> Combining deep learning and Gaussian processes for calibrating a water balance model
Youngsun Park - <i>Colorado State University</i> Which Air Pollutants Spike Together? A Clustering Method for Multivariate Extremes with a Mathematical Foundation
Michael Shull - <i>Penn State</i> Stopover Path Selection Functions for Jointly Modeling Individual and Population-level Movement of Migratory Birds
Tommy Tang - <i>Sandia National Labs</i> Multivariate extreme value models for evaluating the impact of compound extreme weather events on the California power grid.
Gihani Vidumini Wickramasinghe Wickramasinghe Disanayakage - <i>Texas Tech University</i> Transmission Network Individual-Level Modeling of Influenza Under Surveillance Data Limitations
Annie Xu - <i>UT Austin</i> Methods for Ecoacoustic Data: Applications in Tropical Biodiversity and Bird Migration
Tianlu Xue - <i>New Jersey Institute of Technology</i> From Geographic Adjacency to Storm Co-occurrence: Graph-Based Models for Spatiotemporal Hail Forecasting
Rose Graves - <i>Rice University</i> Forecasting Red Tide (<i>Karenia Brevis</i>) trajectories using Bayesian Echo-State Networks on the Florida Gulf Coast
Nicole Kamarat - <i>Montana State University</i> Permutation Exploration for Improvement of Scalable Spatial Stream Network Models and Knowledge of Freshwater Habitats
Swapnonil Mondal - <i>Texas A&M University</i> Weighted Bayesian Bootstrap for High-Dimensional Variable Selection

Jose Palacio - *Rice University*

Inferring Spatial Transmission Dynamics of Respiratory Syncytial Virus Across Houston Wastewater Treatment Plants

Jacob Spooner - *Kansas State University*

Evaluating the Predictive Performance of Predictor Distribution Alignment in a Spatial Principal Component Downscaling Model: A South-Central Kansas Precipitation Case Study

Jay Xu - *Boston Children's Hospital & Harvard CAUSALab*

Causal Source Apportionment Paradigm for Environmental Mixture Exposures

Wentao Zhan - *University of Wisconsin-Madison*

Tensor Covariance Estimation via Kronecker-Structured Sparse Inverse Cholesky

Rong Zheng - *New Mexico State University*

Model-based clustering on the spatial and temporal patterns of tornadoes in the US

Zhengyuan Zhu - *Iowa State University*

Predicting Urban Heat Island Intensity via Robust Spatially Clustered Nonlinear Regression
